[Adaptive] Implement a Sentiment Analysis Model on a New Dataset

BeeNeural Machine Learning Task | Irha Fatima | 27 September 2026

Objective

Build and evaluate a binary sentiment classifier on a new text dataset, compare a neural network with a conventional baseline, and save a model for inference.

Dataset and preparation

The UCI Sentiment Labelled Sentences dataset contains 3,000 labeled reviews from Amazon, IMDb and Yelp. Label 0 denotes negative sentiment and 1 positive sentiment. After removing empty and duplicate text and any conflicting duplicate labels, 2,982 unique reviews remained. The random seed was 42. Source-and-label stratification produced 1,908 training, 477 validation and 597 test reviews. TF-IDF word unigrams and bigrams were fit inside the training pipelines to prevent leakage.

Models and selection

The baseline used TF-IDF and logistic regression. The neural candidate used TF-IDF and a multilayer perceptron with one hidden layer of 64 units, regularization and early stopping. Validation F1 was specified in advance as the selection metric. The higher-scoring architecture was retrained on training plus validation data and evaluated on the held-out test set once.

Model / split	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic / validation	79.45%	81.90%	75.73%	78.70%	0.881
MLP / validation	77.78%	85.19%	67.36%	75.23%	0.855
Selected logistic / test	82.41%	81.09%	84.62%	82.82%	0.894

Interpretation

The logistic regression baseline achieved the higher validation F1 (78.70% versus 75.23%) and was therefore selected. Its test accuracy was 82.41%, with 82.82% F1 and 0.894 ROC-AUC. On 597 test reviews, the positive class had precision 81.09% and recall 84.62%; the negative class had precision 83.86% and recall 80.20%. The neural network was implemented and evaluated, but it was not the strongest final classifier in this experiment.

Limitations and reproducibility

This is a small English-language dataset with only three review domains. Results may not transfer to longer texts, sarcasm or other domains. TF-IDF does not preserve full word order. The repository contains the training and inference scripts, requirements, random seed and instructions. The run writes metrics.json, a confusion matrix, test predictions and a saved inference pipeline.

Dataset attribution

Kotzias et al., “From Group to Individual Labels Using Deep Features,” KDD 2015. UCI Machine Learning Repository, Sentiment Labelled Sentences, dataset 331 (CC BY 4.0). https://archive.ics.uci.edu/dataset/331/sentiment+labelled+sentences